Sahaj Bhadja

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Summary

Software engineer with 2+ years developing **Python data pipelines**, **automated QC workflows**, and **cloud-compatible processing systems** for high-volume multimodal datasets with strong reliability and observability practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **ETL-style Python/FastAPI pipelines** for 5K+ monthly jobs with validation, reconciliation, and metadata-aware processing, producing reliable outputs for downstream analysis and improving production workflow resilience.
- Optimized PostgreSQL and Azure SQL data processing paths for 500+ users, reducing latency from **1.2s to 350ms** while maintaining consistent query behavior across concurrent, data-intensive workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Developed **Python benchmarking pipelines** over 100K+ trajectory logs, generating reproducible metrics and diagnostics to evaluate temporal behavior, model robustness, and algorithm performance under varied experimental conditions.
- Collaborated with multidisciplinary researchers to convert analytical findings into implementation plans, improving scientific feedback loops and accelerating delivery of reliable, data-driven model iteration strategies.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built high-throughput **data processing workflows** for 10K+ daily requests, integrating multimodal model outputs, OCR extraction, and structured persistence while preserving consistency of grading and feedback data quality.
- Implemented **automated testing and CI/CD checks** for model-backed services, reducing turnaround time by **60%** and improving confidence in production releases through validation-first deployment practices.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **cloud data pipelines** with AWS Glue, S3, and Python/Java services, improving response times by **30%** while enabling robust cross-system data movement for distributed production environments.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito, increasing CI efficiency by **68%** and reducing technical debt by **20%** through continuous verification and defect prevention.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Developed REST and WebSocket modules supporting 2M+ synchronized records across eight workflows, improving reliability of large data updates and consistency of operational state transitions in production-like environments.
- Automated Docker/Jenkins test and deployment workflows to reduce release time by **50%**, improving reproducibility of builds and enabling faster, safer iterations across collaborative engineering teams.

Skills

Languages: **Python**, Java, SQL, JavaScript/TypeScript, C++
Data Engineering: ETL workflows, metadata-aware processing, timeseries-style evaluation, algorithm benchmarking
Cloud/Compute: AWS (S3, Glue, RDS), Azure SQL, Docker, Kubernetes, Linux
Quality/Automation: Git, Jenkins, GitHub Actions, CI/CD, PyTest, Mockito, validation pipelines
Collaboration: cross-functional research/engineering execution, technical documentation, agile delivery

Projects

CaptionCraft - Image Caption Generator | *Python, Transformers, Vision Models*

- Benchmarked BLIP, GIT, and ViT-GPT2 on a 10K-image and 100K-caption dataset, improving caption accuracy from **81% to 96%** through model evaluation, tuning, and repeatable inference pipeline design.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Built a **containerized data API** with automated validation checks, CI/CD release controls, and cloud deployment workflows, achieving **87%** processing accuracy with stable, versioned execution in production environments.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Implemented schema-aware query generation for 10+ databases with structured validation and execution outputs, improving reliability of large dataset exploration and reducing manual troubleshooting in analytical data workflows.